Milestone Three: Enhancement Two: Algorithms and Data Structures Narrative

Antoine Boylston

Bachelor of Science in Computer Science, SNHU

CS 499: Computer Science Capstone 2026

Instructor: Alan Spencer, M.S.

March 26, 2026

Narrative

The artifact selected for this enhancement is a **text-based adventure game written in Python**, titled *Dragon King Assassin*, which I originally created in an early programming course. The game allows a player to navigate between rooms, collect items, and defeat a final boss depending on whether the required items have been collected. While the original version of the game successfully demonstrated basic programming concepts such as loops, conditionals, and dictionaries, it did not fully utilize algorithmic problem-solving techniques or advanced data structure concepts. This made it an ideal candidate for enhancement in the algorithms and data structures category.

I selected this artifact for inclusion in my ePortfolio because it provides a clear opportunity to demonstrate growth from basic data handling to more advanced algorithmic design. In the original implementation, the room layout was stored in a nested dictionary and used only for direct lookups when the player moved between rooms. In the enhanced version, I transformed this structure into an explicit **graph representation**, where each room acts as a node and each directional connection represents an edge. I then implemented a **breadth-first search (BFS)** algorithm to compute the shortest path between rooms. This algorithm is used to power a new "Hint" feature that recommends the next optimal move based on the player's current state. The system first identifies rooms containing uncollected items and calculates the shortest path to the nearest one. Once all items are collected, the algorithm instead computes the shortest path to the final boss room. This enhancement demonstrates my ability to apply algorithmic principles, use appropriate data structures such as adjacency lists and queues, and implement traversal logic

to solve a practical problem. Additional improvements included adding test cases to validate BFS behavior and ensuring the system could handle multiple valid shortest paths.

The enhancements made to this artifact align strongly with the course outcomes identified in Module One. Most notably, I achieved **Outcome 3**, which focuses on designing and evaluating computing solutions using algorithmic principles and managing design trade-offs. By converting the room structure into a graph and implementing BFS, I demonstrated the ability to move beyond simple data storage and apply algorithmic reasoning to improve functionality. I also supported **Outcome 4** by implementing a meaningful feature (the Hint system) that delivers real value to the user and reflects industry-relevant problem-solving techniques. At this time, I do not need to update my outcome-coverage plans, as the enhancements align well with the goals I originally established for this category.

Reflecting on this process, I learned how to translate a conceptual data structure into a working implementation and apply an algorithm in a way that meaningfully enhances an application. One of the most valuable takeaways was understanding how existing data can be restructured into a graph to unlock new capabilities without completely redesigning the system. A key challenge I encountered was ensuring the BFS implementation correctly tracked visited nodes and reconstructed the path in the correct order. Additionally, I had to adjust my testing approach to account for multiple valid shortest paths rather than assuming a single correct output. This required me to think more carefully about how to validate algorithmic correctness. Overall, this enhancement strengthened my understanding of graph traversal, algorithm design, and how data structures directly impact the capabilities and scalability of a program.

Prompts used (AI Assistance Disclosure)

Generative AI (ChatGPT) was used to assist with refining the clarity, organization, and academic tone. AI assistance was used only for writing support and editing suggestions. All ideas, reflections, and final content decisions were reviewed and approved by the author.

Sample prompt included:

- “Polish this section for clarity and academic tone”
- “Please check this doc for major errors”

All final content was reviewed and edited by the author.